Offshore–nearshore wave dependence across contrasting NSW coastal settings: an information-theoretic analysis

1. Motivation, Aim and Related Work

1.1 Motivation

Nearshore wave conditions are important for coastal erosion, inundation, sediment transport and hazard assessment, but they cannot always be inferred directly from offshore conditions. As waves move into shallow water, processes such as shoaling, refraction, breaking and headland sheltering modify their characteristics. Because these transformations depend on coastline orientation, embayment geometry and nearshore bathymetry, similar offshore waves can produce different nearshore responses at different coastal settings.

1.2 Related work

Kinsela et al. (2024) compared offshore–nearshore significant wave height at Sydney–Stockton, Wollongong (Port Kembla) –Figure Eight Pools and Batemans Bay–Broulee. Wave height was attenuated at all sites, but the embayed Stockton and Broulee sites showed greater scatter than the open-coast Figure Eight Pools site, suggesting greater sensitivity to offshore wave period and direction. Figure Eight Pools' weaker attenuation was associated with its open coastline, steep bathymetry and greater deployment depth.

Existing numerical approaches, including the NSW Nearshore Wave Tool (Doyle et al., 2025), likewise transform offshore conditions using wave height, period and direction. Mutual information has also been used in wave research, for example by Zamani et al. (2008) to identify informative variables for wave-height forecasting. However, these studies do not quantify offshore–nearshore wave dependence or the additional information provided by offshore period and direction, nor compare these relationships across contrasting NSW coastal settings. This project addresses that gap using MI and CMI.

1.3 Aim

This project will therefore compare two locations in the Kinsela et al. case study: **Sydney–Stockton**, representing an embayed setting, and **Wollongong–Figure Eight Pools**, representing a more exposed open coast. The central question is: **how do information relationships between offshore and nearshore wave conditions differ between these contrasting coastal geomorphic settings?** Mutual information will quantify the dependence between offshore and nearshore significant wave height, while conditional mutual information will quantify how much additional information offshore peak wave period and direction provide after offshore wave height is known.

1.4 Hypothesis

It is hypothesised that offshore wave height will contain relatively more information about nearshore height at Figure Eight Pools, whereas offshore peak period and direction will

provide greater additional information at Stockton. Any differences will be interpreted as **site-specific contrasts consistent with geomorphic influences**, rather than as evidence that geomorphology alone caused the observed information relationships.

2. Method

2.1 Dataset

Based on two offshore–nearshore buoy pairings examined by Kinsela et al. (2024), this study will analyse **Sydney–Stockton (6 Dec 2019–9 April 2021)** and **Wollongong–Figure Eight Pools (6 Dec 2016–11 Dec 2017)**.

The analysis will use publicly available wave observations from the NSW Ocean Wave Data Collection Program (Manly Hydraulics Laboratory, 2013) and NSW Nearshore Wave Buoy Program (NSW Department of Climate Change, Energy, the Environment and Water, 2023). The principal variables will be offshore significant wave height H_o, nearshore significant wave height H_n, offshore peak wave period $T_{p,o}$, and offshore peak wave direction $\theta_{p,o}$. $T_{p,o}$ and $\theta_{p,o}$ are selected because they characterise the dominant spectral component of the offshore wave field and are therefore directly relevant to nearshore transformation.

2.2 Preprocessing

Wave direction requires special treatment because it is circular: for example, 359° and 1° represent similar directions despite their numerical separation. Offshore direction will therefore be represented by $D_{p,o}$= ($\cos\theta_{p,o}$, $\sin\theta_{p,o}$) and treated as a two-dimensional

directional variable.

Nearshore observations will initially be matched to corresponding offshore timestamps so that the analysis uses concurrent observations.

Measurements failing documented quality-control criteria will be excluded; valid extreme-wave observations will be retained because they represent genuine system behaviour. Missing periods will not be interpolated because interpolation could introduce artificial dependence between the two series. The usable sample size and temporal coverage will be reported separately for each pair.

2.3 Information-theoretic analysis

The analysis will be conducted separately for each buoy pair. Mutual information $I(H_o; H_n)$ will quantify the statistical dependence between offshore and nearshore significant wave height. Mutual information is preferred to Pearson correlation because it can capture nonlinear statistical dependence between offshore and nearshore wave conditions.

Conditional mutual information will then be used to determine whether additional properties of the offshore wave field provide information about nearshore wave height beyond offshore height alone: $I(T_{p,o}; H_n \mid H_o)$ for offshore peak wave period and $I(D_{p,o}; H_n \mid H_o)$ for offshore peak wave direction. These conditional measures therefore test whether offshore peak wave period or direction reduces uncertainty about nearshore wave height once offshore significant wave height H_o is already known.

These continuous variables will be analysed using the **Kraskov k-nearest-neighbour estimator** implemented in Java Information Dynamics Toolkit (JIDT; Lizier, 2014), which

is appropriate for continuous variables and does not assume a linear relationship. The initial neighbourhood parameter will be k=4, with a limited sensitivity analysis using nearby values to determine whether the qualitative conclusions are robust.

Because offshore wave conditions may reach the nearshore site after a propagation delay, a limited sensitivity analysis using a small set of physically plausible temporal lags will be conducted using lagged MI/CMI calculations in JIDT (Lizier, 2014).

The wave records are temporally autocorrelated, so the autocorrelation timescale will first be examined and, where appropriate, JIDT's dynamic-correlation-exclusion (Theiler) window (Lizier, 2014) will be applied to the Kraskov estimator to prevent temporally adjacent observations from dominating nearest-neighbour searches. Statistical significance will be assessed using JIDT's surrogate-based significance testing (Lizier, 2014), while noting that standard permutation surrogates may not fully preserve temporal autocorrelation.

Finally, MI and CMI estimates will be compared between the two case-study settings.

3. References

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4. AI Acknowledgement

I used ChatGPT (OpenAI) to assist with brainstorming and refining the project topic, identifying relevant literature and datasets, checking the clarity and structure of the proposal, and improving the wording of selected sections. I independently reviewed the sources, selected the final research question and methodology, and take responsibility for the final content and analysis.

I also used Grammarly.